IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

EEE PART II: MEng, BEng and ACGI

ANALOGUE ELECTRONICS 2

Wednesday, 10 June 10:00 am

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Q1 carries 40% of the marks. Questions 2 and 3 carry equal marks (30% each).

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : T. Constandinou
Second Marker(s) : C. Toumazou

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1. This question consists of 6 sub-questions. Each item can be answered in a few words or a short paragraph with use of a diagram and/or expression where helpful. Please answer all sub-questions and simplify expressions where possible stating any assumptions made.

   a) Explain why it may be preferable to implement a single transistor amplifier using a NPN device instead of using a PNP device.

   [5]

   b) The frequency response of an operational amplifier can be described in terms of its -3dB bandwidth, unity gain bandwidth and open loop gain. Given a unity gain bandwidth of 10MHz and open loop gain of 40dB, what is expected to be the gain at a frequency of 10kHz?

   [5]

   c) Derive expressions (by inspection) for the voltage gains of the amplifier circuits shown below in Fig. 1.1. Include the effect of a finite output resistance.

   [10]

(i)

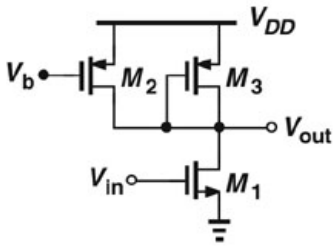


Fig. 1.1(a)

(ii)

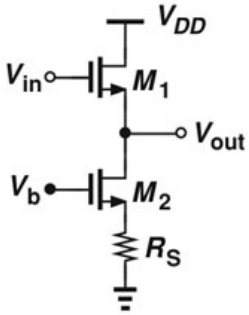


Fig. 1.1(b)

   d) The differentiator shown below in Fig. 1.2 is used to amplify a sinusoidal input at a frequency of 1MHz with a gain of 4. If $A_0=\infty$, determine the value of R_1C_1.

   [5]

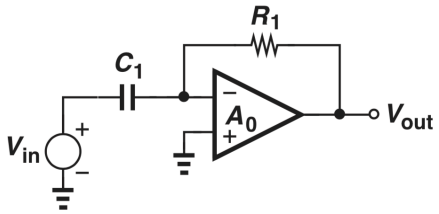


Fig. 1.2

   e) Assuming $V_A<\infty$, compute the differential voltage gain of each stage shown below in Fig. 1.3. (Hint: use the half circuit approximation).

   [5]

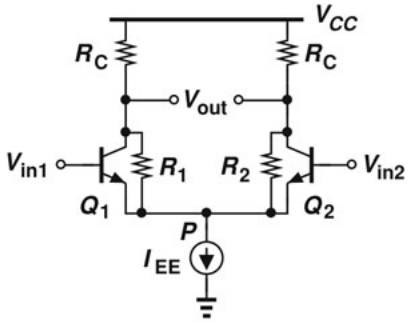


Fig. 1.3

f) An adventurous student designs the following circuit shown in Fig. 1.4. Unfortunately, the amplifier does not operate well.

(i) Prove by inspection that the feedback is positive; [5]

(ii) Breaking the loop at the gate of M_2, determine the loop gain and prove that the feedback is positive. [5]

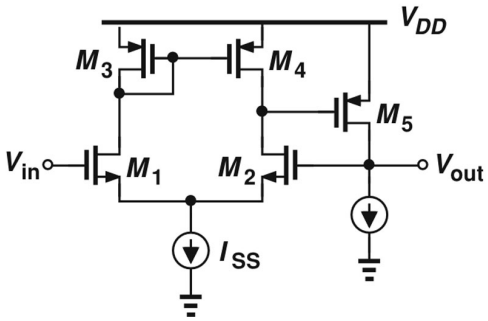


Fig. 1.4

2. The circuit shown below in Fig. 2.1. is a two stage amplifier, specifically a common source stage followed by a source follower stage.

Assume all devices are in saturation and $\lambda>0$ (i.e. $R_{out}<\infty$).

Where required, use the following values:

$\mu_n C_{ox}=200\mu A/V^2$	$V_{TH,n}=0.3V$	$\lambda_n=0.1V^{-1}$
$C_{GS}=(2/3)WLC_{ox}$	$C_{ox}=10fF/\mu m^2$	$C_{DB}\approx 0$
$C_{GD}=C_0W$	$C_0=0.5fF/\mu m$	$C_{SB}\approx 0$

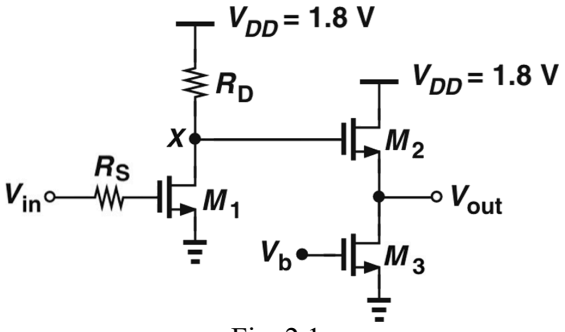


Fig. 2.1.

a) Determine an expression for the low frequency voltage gains of the two stages, i.e. (V_x/V_{in}) and (V_{out}/V_x), and hence the overall low frequency voltage gain (V_{out}/V_{in}).

[5]

b) Evaluate the low frequency voltage gains, given $(W/L)_1=(W/L)_2=(W/L)_3=10\mu m/0.18\mu m$, $R_D=2k\Omega$, and $I_{D1}=I_{D2}=I_{D3}=0.5mA$.

[5]

c) Redraw the circuit above to: (i) identify the 3 nodes that are associated with poles and (ii) include all the parasitic device capacitances (within $M_{1\text{-}3}$).

[5]

d) Determine expressions for the capacitances (to ground) at each of the nodes identified above by lumping together the parasitic capacitances appearing at each node. Use Miller's theorem to resolve any floating (i.e. ungrounded) capacitances.

[5]

e) Determine expressions for the two pole frequencies (excluding the output pole) and evaluate given $R_S=8k\Omega$.

[5]

f) Comment on the high frequency output impedance of the circuit.

[5]

3. The circuit shown below in Fig. 3.1. is called a *Operational Transconductance Amplifier (OTA).*

The circuit operates as follows: Device M_9 provides the bias current (sink) to the differential pair (M_1, M_2). The two tail currents (I_{D1}, I_{D2}) are then copied by the active current mirrors (M_3, M_4), (M_5, M_6) and (M_7, M_8) such as to subtract current I_{D8} from I_{D6} at the output node. This output current (i.e. I_{D6}-I_{D8}) is typically derived to an external load. Mirrors (M_5, M_6) and (M_7, M_8) can be scaled 1:K to provide additional current gain.

Assume all devices are in saturation and $\lambda>0$ (i.e. $R_{out}<\infty$).

Where required, use the following values: $\mu_n C_{ox}=250\mu A/V^2$ $V_{TH,n}=0.4V$ $\lambda_n=0.1V^{-1}$
$\mu_p C_{ox}=100\mu A/V^2$ $V_{TH,p}=-0.55V$ $\lambda_p=0.25V^{-1}$

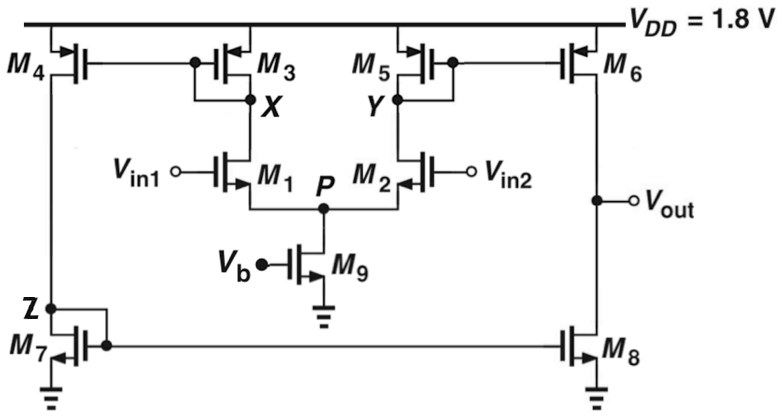


Fig. 3.1.

a) This OTA can be used as a voltage amplifier, provided that the load resistance is large.

Determine an expression for the voltage gain of the OTA, i.e. $A_v=V_{out}/(V_{in2}-V_{in1})$ given that $(W/L)_6=K(W/L)_5$ and $(W/L)_8=K(W/L)_7$.

[5]

b) Given that $I_{D9}=100\mu A$, $(W/L)_{1,2}=100/3$, $(W/L)_{3,4,5,7,9}=3/3$, K=5, evaluate the Voltage Gain and Power Consumption.

[5]

c) How would any variation in the bias current impact the voltage gain? In practice why is it challenging to make stable, well-defined bias currents on-chip?

[5]

d) Comment on this amplifiers expected high frequency performance compared to a typical two stage op-amp (i.e. 1st stage comprising of differential pair with active current mirror load, 2nd stage common source amplifier).

[5]

e) Describe how the circuit can be modified to:

(i) Increase its common mode rejection ratio; [5]

(ii) Reduce its output resistance. [5]